To the Editors of *Computer Physics Communications*,

Dear Editors,

We submit for consideration as a Computer Programs in Physics (CPiP) article the manuscript **"Certifying neural eigenpairs and degenerate eigenspaces of the Schrödinger equation: a self-verifying a posteriori protocol and its implementation"** by Chunyang Chen and Yuhao Peng (equal contribution), School of Mathematics and Physics, China University of Geosciences (Wuhan).

Neural-network solvers of the stationary Schrödinger equation are normally evaluated by comparing with a reference solution. In a genuinely new problem that reference does not exist, and a small residual does not bound the error of the eigenfunction: in our own runs a residual of $4\times10^{-12}$ belonged to one arbitrary member of a degenerate pair, and independent runs disagreed about which member they had found (pairwise overlaps of 0.01). The manuscript turns this evaluation into a computation. It presents a protocol, and a public implementation, that takes a network output on a finite-difference grid and returns (i) an interval *verified* to contain exactly one eigenvalue of the discrete operator, (ii) a two-sided enclosure of that eigenvalue, (iii) a bound on the angle to the exact eigenfunction, and (iv) a verified state index. The hypotheses of the classical Kato–Temple and Davis–Kahan theorems are decided by counting the spectrum with Sturm sequences (1D) and with the Sylvester inertia of a no-pivot sparse LU factorization (2D), with guards against silent pivoting and a resolution floor below which the interval is refused.

The part we believe is new is not an inequality but a construction: because a degenerate level defines no single eigenfunction, the certificate is extended from states to invariant subspaces. The multiplicity is *measured* rather than assumed, the cluster receives a two-sided enclosure at the counting resolution, and a principal-angle bound certifies the subspace that a set of network outputs spans, without any hypothesis on their mutual separation. Using it we can decide, for four independently trained networks on the same degenerate level, which pairs actually span that level: three had collapsed onto the same symmetry-broken representative. The certificate refuses their pair, accepts every pair that contains the orthogonal partner with bounds of 5.2–$6.3\times10^{-3}$, and prices the ambiguous pair—one that omits the partner but whose second direction is mostly noise—at 0.68, so that the weakness is visible in the number rather than hidden. A controlled splitting of the level shows that the boundary between individual and subspace certification sits at the counting resolution, not at degeneracy: in our scan the state certificate refuses at 0.95 times the resolution and accepts at 1.01 times it. Guaranteed cluster bounds for conforming discretizations exist (Cancès et al., Math. Comp. 2020); the construction here does not assume Galerkin structure, which is what allows it to apply to a network output.

Two further constructions close the caveats a referee would otherwise raise. First, the count itself is re-derived in directed-rounding interval arithmetic: the one-dimensional Sturm recurrence, and the two-dimensional count whenever the potential separates as Vx + Vy (where the operator is a Kronecker sum and the count is a comparison of eigenvalue sums), are evaluated with every rounding forced outward, so that a certified count is a theorem about the stored matrix and an unresolved pivot is refused by index rather than rounded to an integer; bisecting on the certified count yields certified brackets for the discrete eigenvalues. The general two-dimensional interval elimination is measured, reported and *not* used, because it refuses on most shifts. Second, because every statement above concerns the discretization, the manuscript adds a bound on the continuous operator: the interpolant of a sampled eigenfunction—a network's included—is a conforming trial function, so its Rayleigh quotient bounds the continuous eigenvalue from above with no reference solution and no interpolation constant. The matching lower bound needs a comparison principle; we give it for the infinite well and the box, and we show it *fails* for the finite well, where a certified discrete energy can lie above the continuous one because the stencil is not variationally consistent at a jump.

The audit is backed by references computed without the certificate: 39 of 39 accepted enclosures contain a

discrete eigenvalue obtained independently—a dense LAPACK solve in two dimensions, never the ARPACK values that propose the intervals—and the counting is cross-checked against dense diagonalization in both dimensions.

The submission includes a 143-test suite, the drivers that regenerate every table and figure of the article, and an appendix recording three reproducible training failures—each with its saved run—that make a network look fine while it cannot be certified. The package is released under the MIT licence, which is on the journal's list of approved licences, and its Program Summary follows the CPiP template; the source archive, a README describing the directory structure, and a test run are uploaded with the manuscript. We state the limitations explicitly: every two-sided enclosure refers to the discretized operator, and the bridge to the continuous operator is one-sided; the interval count covers one dimension and separable two-dimensional potentials, while the general two-dimensional count remains floating point with guards; and the protocol does not apply when the discrete operator cannot be assembled.

The manuscript is not under consideration elsewhere and has not been published. We do not suggest referees. The authors declare no competing interests. The code, the drivers that regenerate every table and figure, and the 143-test suite are available at https://github.com/YuhaoPeng1103/PINN_equation (MIT licence).

Yours sincerely,

Yuhao Peng (corresponding author), on behalf of both authors
School of Mathematics and Physics, China University of Geosciences (Wuhan)
pngyuo@cug.edu.cn